

Program : Diploma in Polymer Technology	
Course Code : 5093B	Course Title: Speciality Polymers
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4,T:0,P:0)	Periods per semester: 60

Course Objectives:

- To impart knowledge on - high performance and advanced technological applications of polymers.

Course Prerequisites:

Topic	Course code	Course name	Semester
General idea about different type of polymerization chemistry and technique		Fundamentals of Polymer Science	3
General idea about elastomers		Synthetic rubber	3
General idea about plastics		Plastic Technology	4

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Explain the structure, properties and application of high temperature and fire-resistant polymers.	15	Understanding
CO2	Describe the structure, properties and application of polymers used in electrical and electronics fields.	15	Understanding
CO3	Explain the properties and application of polymers used in construction, biomedical, aerospace and telecommunication	14	Understanding
CO4	Describe the properties and application of ionomers, hydrogel, nano materials and nano composites.	14	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the structure, properties and application of high temperature and fire-resistant polymers		
M1.01	Generalize the factors affecting the thermal stability and fire resistance of polymers.	2	Remembering
M1.02	Explain methods for improving the thermal stability and flame resistance of polymers.	2	Remembering
M1.03	Describe the properties and high temperature applications of polymers-PTFE, Aromatic Polymers -Poly phenylene Sulphide, PPO.	6	Understanding
M1.04	Explain the properties and applications of PEEK, Aramid fibres, Heterocyclic polymers-PBI	5	Understanding
Contents: High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers – LOI- Definition Properties and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI)			
CO2	Describe the structure, properties and application of polymers used in electrical and electronics fields.		
M2.01	Define the conducting polymers with their conducting mechanism.	2	Remembering
M2.02	Describe the structure, properties and application of conducting polymers like Polyacetylene, poly aniline, poly paraphenylene	5	Understanding

M2.03	Describe the properties and application of photo conducting and piezo electric polymers - Poly vinyl carbazole , PVDF	4	Understanding
M 2.04	Explain the properties and application of Liquid crystalline polymers and Photo resists polymers for semiconductor application.	4	Understanding
	Series Test - I	1	
Contents: Polymers with electrical and electronic properties Conducting polymers -definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semi conductor fabrication - Liquid crystalline polymers.			
CO3	Explain the polymers used in construction, biomedical, aerospace and telecommunication		
M3.01	Explain the polymer concrete, types.	3	Understanding
M3.02	Describe the polymers and its properties used for biomedical application.	5	Understanding
M3.03	Explain the properties of polymers used as binders for rocket propellants.	3	Understanding
M3.04	Explain polymers used for Optical fibre for telecommunication cables (POFC).	3	Understanding
Contents: Polymer concrete- preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) – manufacturing, properties and application			
CO4	Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers and nano materials.		
M4.01	Define the ionic polymers, poly electrolytes and hydrogel.	2	Remembering

M4.02	Describe the Synthesis, physical properties and applications of ionomers based on PS, PTFE	4	Understanding
M4.03	Explain the properties and applications of biodegradable polymers, PLA, Chitosan.	4	Understanding
M4.04	Generalize the idea of nano polymers, preparation, properties and application of nano composites	4	Understanding
	Series Test - II	1	

Contents:

Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes- definition

Synthesis, physical properties and applications of ionomers based on PS, PTFE,

Biodegradable polymers - Properties and applications of PLA, chitosan.

Polymer Nano composite - Definition, properties and application of nano composites

Text / Reference:

T/R	Book Title/Author
T1	H.F.Mark, (Ed), Encyclopedia of polymer Science & Engineering, John Wiley & Sons, New York, 1989.
R1	Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied Science, 1985.
R2	Manas Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel Dekker, New York, 1993
R3	R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd edition, 1998.
R4	Joseph H Koo, Fundamentals properties and applications of polymer nano composites, Cambridge University Press, 2016.

Online Resources:

Sl.No	Website Link
1	https://www.researchgate.net/
2	https://onlinelibrary.wiley.com
3	https://www.ncbi.nlm.nih.gov